

Error-driven learning in morphophonology

Evidence from an experimental study

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07.12.2021

Overview

1. Theoretical background
2. The present study
3. Learning simulation
4. Behavioral experiment
5. Discussion
6. Conclusion

Theoretical background

Preliminary question

How do children acquire their native language and become competent language users?

Theoretical background

Preliminary question

~~How do children acquire their native language and become competent language users?~~ Okay, let's be more specific:

- ▶ How do children learn to assign concrete meaning to specific phonological forms?
- ▶ e.g. German *-chen* [çɛn] → DIMINUTIVE
 - ▶ *Türchen* [tyːrɛçɛn] “small door”
- ▶ German speaking children have to associate said form with both SINGULAR and PLURAL
- ▶ Often times the stem vowel is additionally changed
 - ▶ *hund* [hʊnd̥] “dog” → *Hündchen* [hynd̥çɛn] “doggy”
- ▶ At initial stage, words are stored as unanalyzed chunks (Penke, 2012)

Theoretical background

Distributional learning

- ▶ Children and adults are statistical learners
- ▶ Information about summary statistics (mean, mode, variance) (Maye, Werker, and Gerken, 2002), relative positioning and order of elements (Aslin, Saffran, and Newport, 1998; Saffran, Aslin, and Newport, 1996) and many more (distributional) regularities are computed
- ▶ Each input is assumed to contribute equally to the “to-be-acquired representations”

Theoretical background

Distributional learning

Problem: Various findings in language learning studies are hard to reconcile based on input statistics alone – e.g. Ramscar et al. (2010) for semantic category, Ellis and Sagarra (2010) for morphology, Harmon, Idemaru, and Kapatsinski (2019), Nixon (2020), and Olejarczuk, Kapatsinski, and Baayen (2018) for phonology

- ▶ Instead, findings are better explainable by **error-driven learning** mechanisms

Theoretical background

Error-driven learning

Error-driven learning

Learning is based on prediction and prediction error

- ▶ Two consequences:
 1. Learning will only occur in case prediction error is diminished
 2. Minimizing uncertainty is the end goal of learning
- ▶ How is this achieved?

Theoretical background

Error-driven learning



Figure 1: Conditioning “Pavlovian dog”

Theoretical background

Error-driven learning

- ▶ *Cue* = conditioned stimulus (*Bell*)
- ▶ *Outcome* = unconditioned stimulus (*Food*)
- ▶ In learning, all *Cues* within an environment compete for their share of **association strength** /→/ to an *Outcome*
- ▶ For comprehension: *Cues* (= phonological word-forms) predict *Outcomes* (= semantics/morphological category)

Theoretical background

Error-driven learning

Distributional learning

Based on *Contiguity*: How often two (or more) stimuli **co-occur** is crucial in learning

Error-driven learning

Based on *Contingency*: How reliably a stimulus **predicts** another one is crucial in learning

The present study

Research question

Is morphophonological learning based on error-driven learning mechanisms? More specifically. . .

1. Does learning order impact which phonological form is associated with what morphological category?
 2. Does *Unlearning* through negative cue-evidence benefit morphophonological learning?
- Tested using the artificial language learning paradigm (Culbertson, 2012)

The present study

The artificial language

Morphological analysis:

SG	<a>	CVC[a]C
PL	<i>	CVCC[i]
DM	<ić>	CVCC[it _ĉ]
DMPL	<ić> + <i>	CVCC[it _ĉ][i]

Table 1: Morphological pattern

Vowel-Zero alternation:

$$\blacktriangleright \emptyset \rightarrow \left[\begin{array}{c} +\text{syl}, +\text{low} \end{array} \right] / \left[\begin{array}{c} \text{C} \end{array} \right] - \left\{ \begin{array}{c} \text{C} \\ \# \end{array} \right\}$$

No alternation:

$$\blacktriangleright \emptyset \rightarrow \left[\begin{array}{c} \emptyset \end{array} \right] / \left[\begin{array}{c} \text{C} \end{array} \right] - \left\{ \begin{array}{c} \text{C} \\ \text{V} \end{array} \right\}$$

The present study

Stimuli



(a) [be.fan]



(b) [bef.ni]



(a) [bef.ni^ɸ]



(b) [bef.ni.tɕi]



Figure 2: Condition (which form occurs in isolation) and learning order (which form is presented first) are the variables of interest

The present study

DM_Unlearning: PL_first (δ^1)

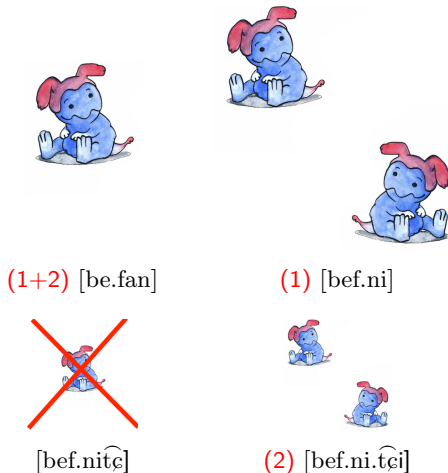


Figure 3: PL was presented in the first phase (32 trials), DMPL second (32 trials); SG as control (32 trials in each phase = 64)

The present study

DM_Unlearning: DMPL_first (δ^2)

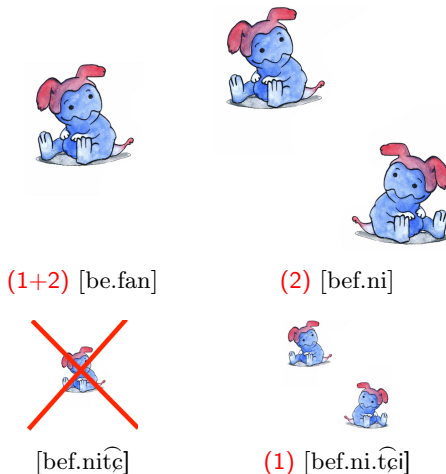


Figure 4: DMPL was presented in the first phase (32 trials), PL second (32 trials); SG as control (32 trials in each phase = 64)

The present study

PL_Unlearning: DM_first (π^1)

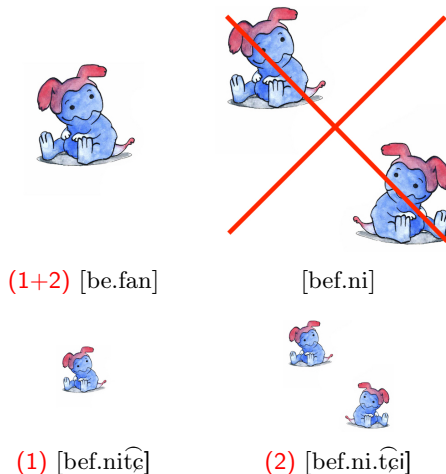


Figure 5: DM was presented in the first phase (32 trials), DMPL second (32 trials); SG as control (32 trials in each phase = 64)

The present study

PL_Unlearning: DMPL_first (π^2)

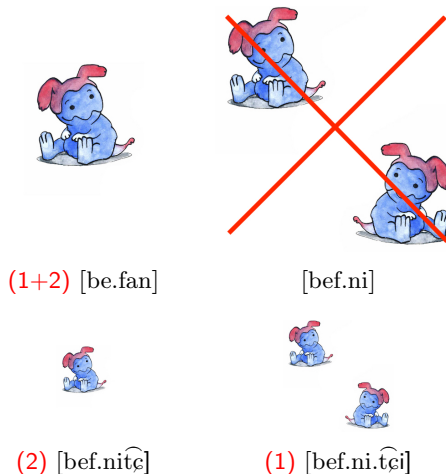


Figure 6: DMPL was presented in the first phase (32 trials) DM second (32 trials); SG as control (32 in each phase = 64)

The present study

Summary design

Abbr.	Group	Learning I	Learning II	Test cues
δ^1	DM:PL_first	$[i] \rightarrow \text{PL}$	$[\widehat{\text{itci}}] \rightarrow \text{DMPL}$	$[\widehat{\text{itc}}] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$
δ^2	DM: DMPL_first	$[\widehat{\text{itci}}] \rightarrow \text{DMPL}$	$[i] \rightarrow \text{PL}$	$[\widehat{\text{itc}}] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$
π^1	PL: DM_first	$[\widehat{\text{itc}}] \rightarrow \text{DM}$	$[\widehat{\text{itci}}] \rightarrow \text{DMPL}$	$[\widehat{\text{itc}}] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$
π^2	PL: DMPL_first	$[\widehat{\text{itci}}] \rightarrow \text{DMPL}$	$[\widehat{\text{itc}}] \rightarrow \text{DM}$	$[\widehat{\text{itc}}] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$

Table 2: Overview of the experimental design

Learning simulation

Rescorla-Wagner learning rule

Error-driven learning can be simulated in a fully connected two-layer (end-to-end) network using the *Rescorla-Wagner* equation (Rescorla and Wagner, 1972)

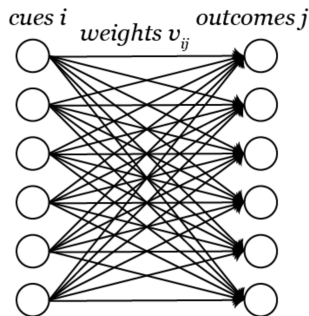


Figure 7: Activation weights between *Cues* and *Outcomes*

Learning simulation

Rescorla-Wagner learning rule

Updating the weight matrix

$$w_{ij}^{t+1} = w_{ij}^t + \Delta w_{ij}^t \quad (1)$$

Rescorla-Wagner equation

$$\Delta w_{ij}^t = \begin{cases} \text{(a) } 0 & \text{if } \textit{Absent}(c_i, t) \\ \text{(b) } \alpha_i \beta_1 \left(\lambda_{j-} \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Present}(o_j, t) \\ \text{(c) } \alpha_i \beta_2 \left(0 - \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Absent}(o_j, t) \\ \text{(d) } 0 & \text{else} \end{cases} \quad (2)$$

Learning simulation

Rescorla-Wagner learning rule

Computational simulations for the four learning groups $\delta^1\delta^2\pi^1\pi^2$ using the `edl-package` (van Rij and Hoppe, 2021)

- ▶ *Cues*: Bigrams of the word forms (computed with the `ndl-package` (Arppe et al., 2018))
 - ▶ e.g. the pseudoword `goplići` was split into its bigrams
 - ▶ `#g, go, op, pl, li, ić, ċi, i#`
- ▶ *Outcomes*: Morphological category (SG, PL, DM)

Learning simulation

Data structure

Computational simulations for the four learning groups $\delta^1\delta^2\pi^1\pi^2$ using the `edl`-package (van Rij and Hoppe, 2021)

- ▶ For DM_Unlearning ($\delta^1\delta^2$): 165 connection weight-values (55 *Cues* x 3 *Outcomes*) for each
 - ▶ Focus on DM-*Outcome*
- ▶ For PL_Unlearning ($\pi^1\pi^2$): 168 connection weight-values (56 *Cues* x 3 *Outcomes*) for each
 - ▶ Focus on PL-*Outcome*

Learning simulation

Results δ^1 (left) vs. δ^2 (right)

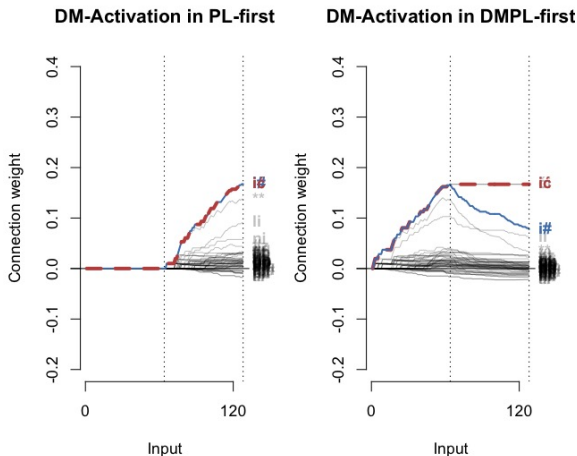


Figure 8: Correct *Cue* iĉ [iĉ] has the strongest activation only in δ^2

Learning simulation

Results π^1 (left) vs. π^2 (right)

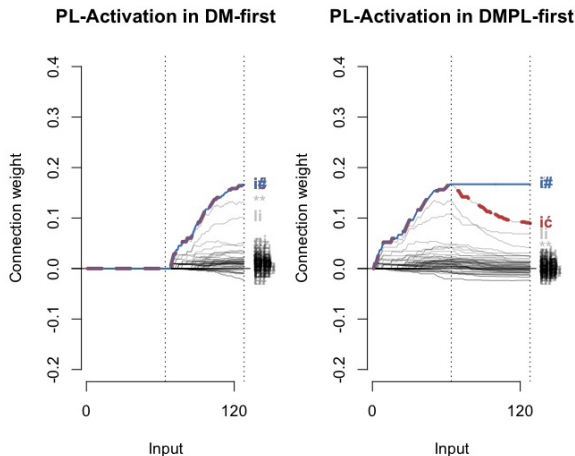


Figure 9: Correct Cue $i\#$ [i] has the strongest activation only in π^2

Learning simulation

Summary simulation

Non-Unlearning	(1) $A \rightarrow X$	(2) $AB \rightarrow XY$	(3) $A \rightarrow Y, B \rightarrow Y$
Unlearning	(1) $AB \rightarrow XY$	(2) $A \rightarrow X$	(3) $A \nrightarrow Y, B \rightarrow Y$

Table 3: Schematic differences between learning order

- Unlearning is only available for the groups presented with DMPL first ($\delta^2\pi^2$)
- Non-Unlearning groups ($\delta^1\pi^1$) are expected to learn the withhold category worse, as the competing, non-discriminative cue has the same activation weight

Behavioral experiment

Hypotheses

1. H_0 : Non-Unlearning- ($\delta^1\pi^1$) and Unlearning-groups ($\delta^2\pi^2$) will not differ in overall correct choice accuracy.
2. H_1 : Overall correct choice accuracy will be higher in the Unlearning-groups due to unlearning – and if this is the case...
3. H_2 : Overall correct choice accuracy for the withhold category (DM in δ^2 , PL in π^2) will be higher due to unlearning.

Behavioral experiment

Participants

116 participants (mean age: 26.91, range 18-69)

Experiments programmed in *PsychoPy* (Peirce et al., 2019) and run online through *Pavlovia* (Bridges et al., 2020)

Behavioral experiment

Procedure

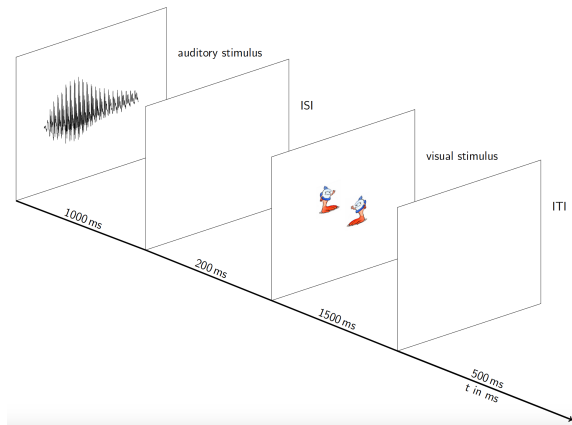


Figure 10: Time course of a single trial for the learning phases I and II (128 in total).

Behavioral experiment

Procedure

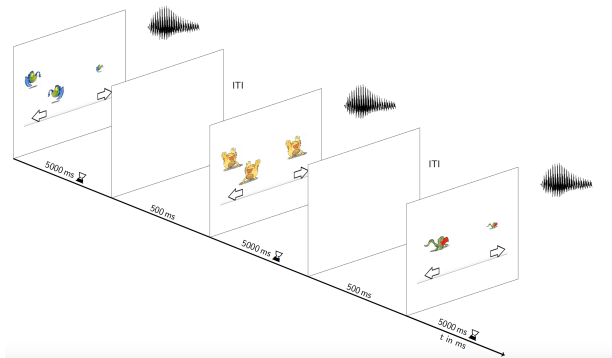


Figure 11: Time course of the test phase (three trials are shown). 64 trials (32 PL, 32 DM)

Behavioral experiment

Results

Overall 7424 data points (116 participants x 64 trials) analyzed in R (R Core Team, 2021)

- ▶ *Generalized linear mixed model*-analysis using `lmer`test (Kuznetsova, Brockhoff, and Christensen, 2017)
- ▶ Random intercepts for Word and Participant
- ▶ Dependent variable: accuracy – 0 (wrong), 1 (right)

Behavioral experiment

Results

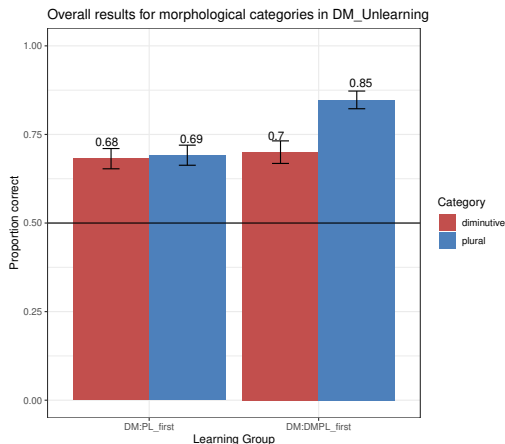


Figure 12: Probability of correct choice for diminutives and plurals in DM_Unlearning (δ^1 and δ^2). Error bars show the 95% confidence interval.

Behavioral experiment

Results

- ▶ Group and Category as fixed effect + Group:Category-interaction
- ▶ No significant effect for Group and Category, but interaction was significant (Est. = 0.9585, SE = 0.1871, $z = 5.123$, $p < 0.001$)
- ▶ Better performance for the PL-category in *Unlearning*-group (no group-differences for withhold DM-category)

Behavioral experiment

Results

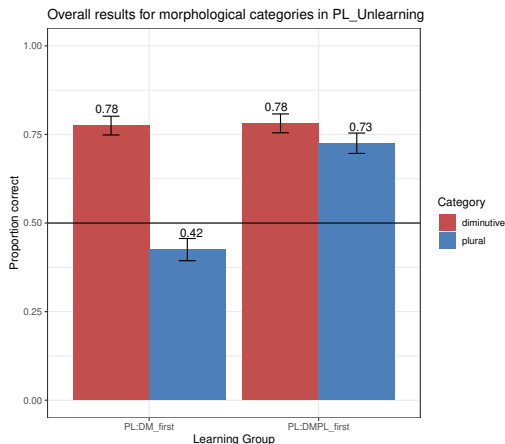


Figure 13: Probability of correct choice for diminutives and plurals in PL_Unlearning (π^1 and π^2). Error bars show the 95% confidence interval.

Behavioral experiment

Results

- ▶ Group and Category as fixed effect + Group:Category-interaction
- ▶ Significant effect of Category (Est. = -2.4817, SE = 0.1643, $z = -15.101$, $p < 0.001$); withhold PL-category was worse in general
- ▶ But: Group:Category-interaction significant (Est. = -2.0623, SE = 0.1916, $z = -10.765$, $p < 0.001$)
- ▶ Significant Category-effect due to PL-accuracy being worse in *Non-Unlearning* only

Discussion

Did *Unlearning* have an effect?

- ▶ Withhold DM-category in DM_Unlearning **did not benefit** from unlearning of the non-discriminative cue
 - ▶ But: Participants performed better in the PL-category instead (85% accuracy vs. 69%)
 - ▶ (H_0 ✗ H_1 ✓ H_2 ✗)
- ▶ Withhold PL-category in PL_Unlearning **did benefit** from unlearning of the non-discriminative cue
 - ▶ Better performance for the PL-category (73% accuracy vs. 42%)
 - ▶ (H_0 ✗ H_1 ✓ H_2 ✓)
- ▶ Learning order and Unlearning affected learning

Discussion

Unlearning only in DM_Unlearning?

- ▶ Unlearning of non-discriminative cue only beneficial if a **more-salient cue** is to be unlearned
- ▶ When $[\text{it}_{\text{ci}}]$ is encountered, $[\text{it}_{\text{ci}}]$ is likely associated as whole form and only at a later stage decomposed
- ▶ Decomposition as consequence of Unlearning

Conclusion

- ▶ Behavioral experiment confirmed learning predictions obtained from edl-simulations
- ▶ Unlearning (and learning order) benefited learning
- ▶ Learning facilitated, if negative evidence and subsequent unlearning are available
- ▶ But: Only when non-discriminative, more salient *Cues* are unlearned
 - ▶ Less salient *Cues* are overshadowed by more salient ones (explaining why the DM-category was still learned, even without Unlearning in DM_Unlearning condition)
- ▶ Error-driven learning mechanisms are present in morphophonological learning

Future outlook

Planned PhD-project

Learning non-concatenative morphophonology

Distributional learning vs. error-driven learning

- ▶ Stay tuned! :)
- ▶ Contact: Erdin.Mujezinovic@hhu.de

That's it!



► Are there any questions? Comments?

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